Snigdha Indupalli

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.01
SRI CHAITANYA JR COLLEGE, MANIKONDA,HYDERABAD	Class XII (TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION)	2024	97.40%
THE FUTURE KID'S SCHOOL, HYDERABAD	Class X (COUNCIL FOR THE INDIAN SCHOOL CERTIFICATE EXAMINATIONS)	2022	91.50%

SCHOLASTIC ACHIEVEMENTS

Qualified JEE Advanced 2024 and secured administration in Indian Institute of Technology Hyderabad

POSITIONS OF RESPONSIBILITY

Public Relations Coordinator, Elan & nVision, IIT Hyderabad

Helped in outreach and public relation initiatives for IIT Hyderabad's annual techno-cultural festival, collaborating with external organizations and media partners.

Outreach & Collaborations Coordinator, TEDxIITHyderabad

Established strategic partnerships and managed outreach efforts with speakers, sponsors, and organizations to support successful TEDx events.

Design Coordinator, Finance and Consulting Club (FCC)

Designed branding and promotional content for club events, ensuring consistent visual identity across digital and print media.

Design Coordinator, Cultural & Literary Council

Developed creative assets and publicity material for institute-wide cultural and literary events, enhancing audience engagement.

Class Representative, IIT Hyderabad

Represented the Engineering Physics batch for the term 2024-25 by coordinating between students and faculty, addressing academic concerns.

SKILLS

Programming Languages: C, C++, Python (NumPy, Pandas, Matplotlib, TensorFlow, Scikit Learn)

Tools & Platforms: Git, GitHub, VS Code, PyCharm, Figma, Canva

Soft Skills: Team Management, Effective Communication, Problem Solving

RELEVANT COURSES

Computer Science and Programming: Introducation to Programming, Data Structures & Applications, Computational Physics

Artifical Intelligence: Introduction to Artificial Intelligence, Introduction to Probability, Introduction to Statistics

Mathematics: Calculus I, Calculus II, Elementary to Linear Algebra, Differential Equations, Maths for Physics, Advanced Mathematical Physics

Electrical & Electronics: Basic Electric Circuits, Analog Electronics, Digital Circuits

Physics: Modern Physics, Classical Mechanics, Special Relativity, Thermodynamics, Fluid Dynamics, Optics and Photonics, Quantum Mechanics

PROJECTS

Hybrid GenAI Log Classification Engine | DeepSeek R1,BERT, Python, Scikit-Learn

- Architected a hybrid NLP pipeline to categorize raw server logs across ~2.4K records, optimizing execution costs and model accuracy
- Engineered and unsupervised text discovery workflow using **BERT** sentence embeddings and **DB scan** clustering to programmatically isolate text patterns
- Developed a cost-efficient routing logic: parsed ~20% of structured logs with **Regex**,mapped high-sample clusters to a localized **Logistic Regression** model, and delegated complex strings to **DeepSeek R1**
- Implemented a probabilistic confidence threshold (<50%) to systematically capture and flag unclassified string exceptions

Spoken Digit Recognition Sytem usig DTW Algorithm

- Developed a system for recognizing spoken digits (0-9) using the **Dynamic Time Warping (DTW)** algorithm. The system analyzes voice input, extracts **Mel Frequency Cepstral Coefficients (MFCCs)** using Python speech features, and matches them against coefficients from pre-recorded samples to identify the spoken digit.
- **TechStack**-Pydub, Speechpy, Numpy, Matplotlib, Scipy.

Multi-Label Comment Toxicity Detector | TensorFlow, Python, Streamlit

- Built a **Bi-directional LSTM RNN** to classify text across 6 toxicity categories using a dataset of ~160K comments.
- Optimized data ingestion using the **tf.data** API (cache,prefetch), eliminating hardware bottlenecks during training.
- Deployed a web application using **Streamlit** to handle real-time text interference and model evaluaion.

EXTRACURRICULAR

Covered conference proceedings through content creation and media documention for the **X Model United Nations** conducted by IIT Hyderabad.

Organized dance workshops, performances and club events for **Suffle Crew-Dance Club of IIT Hyderabad** while contributing to choreography and event coordination.

Represented IIT Hyderabad at the **Inter IIT Cultural Meet** in both Group Dance and Duo Dance categories.

Volunteered in **NSS**, taking part in service activities and campus events, which improved teamwork and social awareness.